Numerical methods

Wiki *Enhanced dissipation for passive scalars*

10 November 2026

Page team: to be named · Status: skeleton, one record
Discuss on Zulip: #numerics, topic "Numerical methods".

Every computed picture and video of this wiki has a record on this page: what was computed, how, with which parameters and checks, and the code that produced it.

1 How a record works

- A record is a section with an anchor `num-...`; the picture or video links to it with the parameter `record` of Media.
- It gives the equation, the parameters, the method, the checks, the code and how to run it, and its status.
- The code is in the folder `numerics/` of the repository of the wiki, one folder per record, with the output of its checks.

2 Records

2.1 Heat versus shear on the torus

Produces the video on the Overview (`heat-vs-shear.mp4`) and the check figure below

Equation $\partial_t f + \sin y\, \partial_x f = \nu \Delta f$ on $[0, 2\pi)^2$, periodic, and the heat equation $\partial_t f = \nu \Delta f$; same datum $f_0 = \cos x$

Parameters $\nu = 10^{-3}$, times $0 \le t \le 100$; video of 20 s at 24 frames per second, time running uniformly (5 time units per second), last frame held 1.5 s

Method one Fourier mode in x: $f = \mathrm{Re}(g(t,y)e^{ix})$ with $\partial_t g = -i \sin y\, g + \nu(\partial_y^2 - 1)g$; 512 Fourier modes in y; Strang splitting with time step 0.01, transport and diffusion steps solved exactly; heat equation solved exactly

Checks with 1024 modes and time step 0.005 the norms agree to a relative $6 \cdot 10^{-8}$; the energy balance $\|g(t)\|^2 + 2\nu \int_0^t \|(\partial_y g, g)\|^2 ds = \|g(0)\|^2$ holds to $3 \cdot 10^{-7}$; decay rate of the norm on $[50, 100]$: 0.0167, against $\nu^{1/2} \approx 0.0316$

Code numerics/heat-vs-shear/heat_vs_shear.py in the repository, also below (Python 3, NumPy, Matplotlib, ffmpeg); `python3 heat_vs_shear.py -out ../../wiki/media`, about a minute

Status computed by Claude (AI) on 8 November 2026; not yet checked by a human [demo text]

Code.

```
#!/usr/bin/env python3
"""Heat equation versus advection-diffusion by a shear flow on the torus [0, 2pi)^2.
```

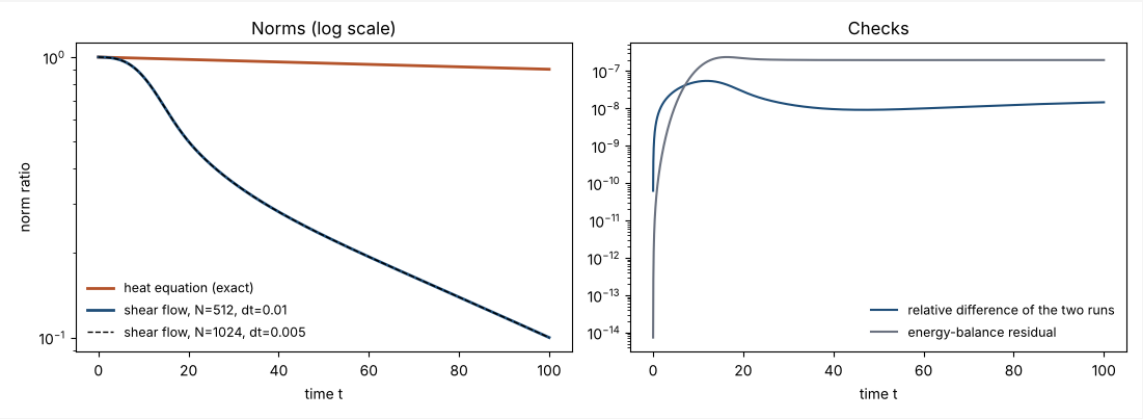


Figure. Left: the norms of the two runs and of the heat equation. Right: the relative difference of the two runs and the energy-balance residual.

```
    d_t f + sin(y) d_x f = nu (d_xx + d_yy) f,      f(0, x, y) = cos x,

compared with the heat equation (the same without the transport term), whose solution is
exp(-nu t) cos x. The initial datum has zero average in x. It is a single Fourier mode in
    x,
so f(t, x, y) = Re(g(t, y) e^{ix}) with

    d_t g = -i sin(y) g + nu (d_yy - 1) g,      g(0, y) = 1,

solved by a Fourier method in y with Strang splitting: the transport step is the exact
multiplication by exp(-i sin(y) dt), the diffusion step is exact in Fourier variables.
The L2 norm reported is ||f(t)|| / ||f(0)||, that is the root mean square of |g(t, .)|.

Checks (written to heat-vs-shear-check.txt and heat-vs-shear-check.png):
  * the run is repeated with twice the Fourier modes and half the time step, and the norms
    are compared;
  * energy balance: ||g(t)||^2 + 2 nu int_0^t ||(d_y g, g)||^2 ds = ||g(0)||^2 (means over
    y).

The video lasts --duration seconds; frame s in [0, 1] shows t = T (a s + (1 - a) s^2), a =
    --slow (default 1:
time runs uniformly; a < 1 slows down the start); the last frame is held --hold seconds.

Outputs, in --out: heat-vs-shear.mp4 (video), heat-vs-shear-poster.png (one frame of it),
heat-vs-shear-check.png and heat-vs-shear-check.txt. Needs NumPy, Matplotlib and ffmpeg
(with the libx264 encoder). Run:  python3 heat_vs_shear.py --out ../../wiki/media

Drafted by Claude (AI) on 8 November 2026 for the wiki "Enhanced dissipation for passive
scalars"; record: page Numerical methods, section num-heat-vs-shear.
"""
import argparse
import os
import shutil
import subprocess
import tempfile

import numpy as np
import matplotlib

matplotlib.use("Agg")
```

```
import matplotlib.pyplot as plt

HEAT_COLOR = "#b4532a"
SHEAR_COLOR = "#1f4e79"


def solve(nu, T, N, dt, frame_times=()):
    """Times, norms ||f(t)||/||f(0)||, energy-balance residuals, and g(t, y) at
    frame_times."""
    y = 2 * np.pi * np.arange(N) / N
    eta = np.fft.fftfreq(N, d=1.0 / N)              # integer wavenumbers in y
    symbol = eta ** 2 + 1.0                          # symbol of -(d_yy - 1): mode k = 1
    in x
    half = np.exp(-nu * symbol * dt / 2)
    transport = np.exp(-1j * np.sin(y) * dt)
    ghat = np.fft.fft(np.ones(N, dtype=complex))
    steps = int(round(T / dt))
    times = dt * np.arange(steps + 1)
    sq = np.empty(steps + 1)                         # mean over y of |g|^2
    dissip = np.empty(steps + 1)                     # mean over y of |d_y g|^2 + |g|^2
    frames, targets, j = [], list(frame_times), 0
    for n in range(steps + 1):
        power = np.abs(ghat) ** 2 / N ** 2           # Parseval: means over y
        sq[n] = power.sum()
        dissip[n] = (symbol * power).sum()
        while j < len(targets) and targets[j] <= times[n] + dt / 2:
            frames.append(np.fft.ifft(ghat))
            j += 1
        if n == steps:
            break
        ghat *= half
        ghat = np.fft.fft(np.fft.ifft(ghat) * transport) * half
    integral = np.concatenate([[0.0], np.cumsum((dissip[1:] + dissip[:-1]) / 2) * dt])
    residual = np.abs(sq + 2 * nu * integral - 1.0)
    return times, np.sqrt(sq), residual, frames, y


def field(g, nx):
    """f(x, y) = Re(g(y) e^{ix}) on an nx-by-nx grid (rows: y, columns: x)."""
    step = max(1, len(g) // nx)
    x = 2 * np.pi * np.arange(nx) / nx
    return np.real(g[::step][:, None] * np.exp(1j * x)[None, :])


def draw_frame(path, t, g, nu, times, norms, heat, nx, ymin):
    x = 2 * np.pi * np.arange(nx) / nx
    f_heat = np.exp(-nu * t) * np.cos(x)[None, :] * np.ones((nx, 1))
    f_shear = field(g, nx)
    fig = plt.figure(figsize=(11.2, 7.2), dpi=100)
    gs = fig.add_gridspec(2, 2, height_ratios=[3.6, 1.15], hspace=0.3, wspace=0.06,
                          left=0.08, right=0.97, top=0.845, bottom=0.095)
    for col, (data, title) in enumerate([(f_heat, "Heat equation:  $u = 0$"),
                                         (f_shear, r"Shear flow:  $u = (\sin y,\, 0)$")]):
        ax = fig.add_subplot(gs[0, col])
        ax.imshow(data, origin="lower", extent=[0, 2 * np.pi, 0, 2 * np.pi], cmap="RdBu_r"
    ,
                  vmin=-1, vmax=1, interpolation="bilinear")
        ax.set_title(title, fontsize=17, pad=8)
        ax.set_xticks([]); ax.set_yticks([])
        ax.set_aspect("equal")
    fig.suptitle(r"Same initial datum $f_0 = \cos x$, same diffusivity $\nu = 10^{-3}$"
                 + "        t = %d" % round(t), fontsize=17, y=0.975)
```

```
    ax = fig.add_subplot(gs[1, :])
    k = np.searchsorted(times, t, side="right")
    ax.semilogy(times, heat, color=HEAT_COLOR, alpha=0.18, lw=2)
    ax.semilogy(times, norms, color=SHEAR_COLOR, alpha=0.18, lw=2)
    ax.semilogy(times[:k], heat[:k], color=HEAT_COLOR, lw=2.4, label="heat equation")
    ax.semilogy(times[:k], norms[:k], color=SHEAR_COLOR, lw=2.4, label="shear flow")
    ax.axvline(t, color="#888888", lw=1)
    ax.set_xlim(0, times[-1]); ax.set_ylim(ymin, 1.5)
    ax.set_yticks([10.0 ** e for e in range(int(np.log10(ymin)), 1)])
    ax.set_xlabel("time  t", fontsize=13)
    ax.set_ylabel(r"$\|f(t)\|\,/\,\|f_0\|$", fontsize=14)
    ax.tick_params(labelsize=11)
    ax.legend(loc="lower left", fontsize=12, frameon=False, ncol=2)
    fig.savefig(path)
    plt.close(fig)


def main():
    ap = argparse.ArgumentParser(description=__doc__.split("\n")[0])
    ap.add_argument("--out", default=".", help="folder for the video and the figures")
    ap.add_argument("--nu", type=float, default=1e-3)
    ap.add_argument("--T", type=float, default=100.0)
    ap.add_argument("--N", type=int, default=512, help="Fourier modes in y")
    ap.add_argument("--dt", type=float, default=0.01)
    ap.add_argument("--duration", type=float, default=20.0, help="length of the video in
    seconds")
    ap.add_argument("--fps", type=int, default=24)
    ap.add_argument("--slow", type=float, default=1.0, help="speed of time at the start,
    relative to the mean (1: uniform)")
    ap.add_argument("--hold", type=float, default=1.5, help="seconds the last frame is
    held")
    ap.add_argument("--nx", type=int, default=256, help="pixels per side of each panel")
    a = ap.parse_args()
    os.makedirs(a.out, exist_ok=True)
    s = np.linspace(0, 1, int(round(a.duration * a.fps)))
    frame_times = a.T * (a.slow * s + (1 - a.slow) * s ** 2)
    times, norms, residual, frames, y = solve(a.nu, a.T, a.N, a.dt, frame_times)
    heat = np.exp(-a.nu * times)
    # check: the same run with N doubled and dt halved
    times2, norms2, residual2, _, _ = solve(a.nu, a.T, 2 * a.N, a.dt / 2)
    diff = np.abs(norms2[::2] - norms) / norms2[::2]
    ymin = 10 ** np.floor(np.log10(norms.min()))
    with open(os.path.join(a.out, "heat-vs-shear-check.txt"), "w") as fh:
        fh.write("nu=%g T=%g N=%d dt=%g frames=%d fps=%d\n" % (a.nu, a.T, a.N, a.dt, len(
    frame_times), a.fps))
        fh.write("final norm: shear %.4e, heat %.4e\n" % (norms[-1], heat[-1]))
        fh.write("max relative difference with (2N, dt/2): %.2e\n" % diff.max())
        fh.write("max energy-balance residual: %.2e (2N, dt/2: %.2e)\n" % (residual.max(),
     residual2.max()))
        fh.write("decay rate of the shear norm on [T/2, T]: %.4f; nu^(1/2) = %.4f\n" % (
            np.log(norms[len(norms) // 2] / norms[-1]) / (times[-1] - times[len(times) //
    2]), np.sqrt(a.nu)))
    # check figure
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4), dpi=100)
    ax1.semilogy(times, heat, color=HEAT_COLOR, lw=2, label="heat equation (exact)")
    ax1.semilogy(times, norms, color=SHEAR_COLOR, lw=2, label="shear flow, N=%d, dt=%g" %
    (a.N, a.dt))
    ax1.semilogy(times2, norms2, color="k", lw=1, ls="--", label="shear flow, N=%d, dt=%g"
     % (2 * a.N, a.dt / 2))
    ax1.set_xlabel("time t"); ax1.set_ylabel("norm ratio"); ax1.legend(frameon=False,
    fontsize=9)
    ax1.set_title("Norms (log scale)")
```

```
    ax2.semilogy(times[1:], diff[1:] + 1e-16, color=SHEAR_COLOR, lw=1.5, label="relative
    difference of the two runs")
    ax2.semilogy(times[1:], residual[1:] + 1e-16, color="#6b7280", lw=1.5, label="energy-
    balance residual")
    ax2.set_xlabel("time t"); ax2.legend(frameon=False, fontsize=9); ax2.set_title("Checks
    ")
    fig.tight_layout()
    fig.savefig(os.path.join(a.out, "heat-vs-shear-check.png"))
    plt.close(fig)
    # video
    with tempfile.TemporaryDirectory() as tmp:
        for i, (t, g) in enumerate(zip(frame_times, frames)):
            draw_frame(os.path.join(tmp, "f%04d.png" % i), t, g, a.nu, times, norms, heat,
     a.nx, ymin)
        poster = min(range(len(frame_times)), key=lambda i: abs(frame_times[i] - 0.2 * a.T
    ))
        shutil.copyfile(os.path.join(tmp, "f%04d.png" % poster), os.path.join(a.out, "heat
    -vs-shear-poster.png"))
        subprocess.run(["ffmpeg", "-y", "-loglevel", "error", "-framerate", str(a.fps),
                        "-i", os.path.join(tmp, "f%04d.png"), "-vf", "tpad=stop_mode=clone
    :stop_duration=%g" % a.hold,
                        "-c:v", "libx264", "-pix_fmt", "yuv420p",
                        "-crf", "26", "-preset", "slow", "-movflags", "+faststart",
                        os.path.join(a.out, "heat-vs-shear.mp4")], check=True)
    print(open(os.path.join(a.out, "heat-vs-shear-check.txt")).read().strip())


if __name__ == "__main__":
    main()
```

2.2 Alternating shears with random phases

Produces the second video on the Overview (`alternating-shears.mp4`) and the check figure below

Equation $\partial_t f + u \cdot \nabla f = \nu \Delta f$ on $[0, 2\pi)^2$, periodic, with $f_0 = \cos x$ and the alternating shears of Pierrehumbert's model: $u = (\sin(y + \varphi_n), 0)$ for $6n \le t < 6n + 3$ and $u = (0, \sin(x + \psi_n))$ for $6n + 3 \le t < 6n + 6$, the phases independent and uniform on $[0, 2\pi)$; compared with the steady shear $u = (\sin y, 0)$ and with the heat equation. Both velocity fields have size at most 1.

Parameters $\nu = 10^{-4}$, each shear acting for a time 3, times $0 \le t \le 60$, phases from the random generator of NumPy with seed 2026; video of 20 s at 24 frames per second, time running uniformly (3 time units per second), last frame held 1.5 s

Method Fourier pseudo-spectral method on a 512×512 grid; Strang splitting with time step 0.0125, the transport steps (each row or column shifted, through its Fourier coefficients) and the diffusion steps solved exactly; the steady shear as in the record above

Checks with 1024×1024 modes and time step 0.00625 the norms agree to a relative $1.2 \cdot 10^{-6}$; the energy balance $\|f(t)\|^2 + 2\nu \int_0^t \|\nabla f\|^2 ds = \|f_0\|^2$ holds to $3.5 \cdot 10^{-6}$; time for the norm to halve, for $\nu = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}$: alternating shears 8.8, 16.3, 23.9, 31.7 (the same with twice the modes), an increase of about 7.6 for each factor 10 in ν; steady shear 8.6, 20.0, 43.9, 94.9; heat equation $\ln 2/\nu$

Context Exponential mixing for randomly shifted alternating shears is proved by Blumenthal, Coti Zelati and Gvalani (2023), and dissipation on the time scale $|\log \nu|$, with high probability, by Cooperman, Iyer and Son (2025); whether the parameters used here satisfy their hypotheses has not been checked

Code numerics/alternating-shears/alternating_shears.py in the repository (Python 3, NumPy, SciPy, Matplotlib, ffmpeg); `python3 alternating_shears.py -out ../../wiki/media`, about 15 minutes

Status computed by Claude (AI) on 8 November 2026; not yet checked by a human [demo text]

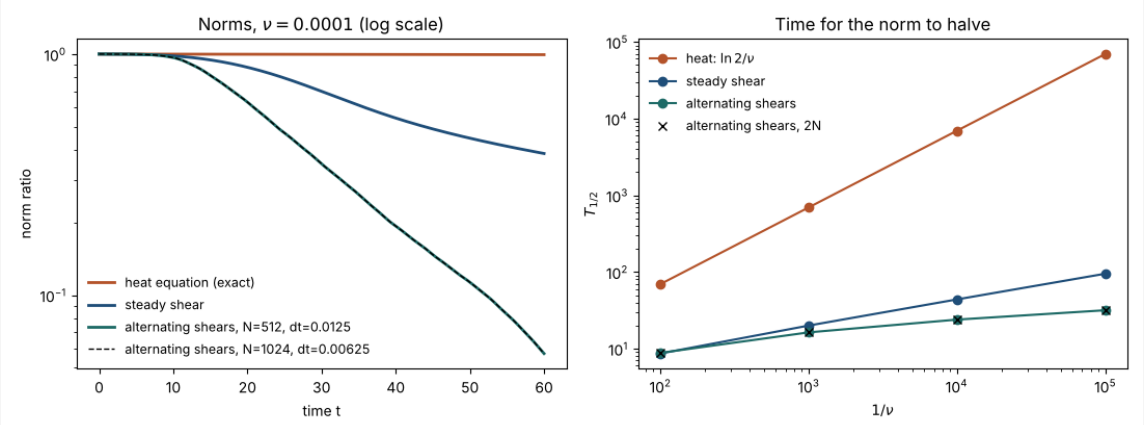


Figure. Left: the norms for $\nu = 10^{-4}$, with two resolutions for the alternating shears. Right: the time for the norm to halve, against $1/\nu$, logarithmic scales.

3 Solvers

[The methods used across the records, with their strengths and limits.]

References

A. Blumenthal, M. Coti Zelati and R. S. Gvalani, *Exponential mixing for random dynamical systems and an example of Pierrehumbert*, Annals of Probability **51** (2023), 1559–1601, arXiv:2204.13651.

W. Cooperman, G. Iyer and S. Son, *A Harris theorem for enhanced dissipation, and an example of Pierrehumbert*, Nonlinearity **38** (2025), 045027, arXiv:2403.19858.